Water Potability Prediction Using Machine Learning

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The Problem

2 billion people worldwide lack access to safe drinking water

Traditional lab testing: slow, expensive, inaccessible

Waterborne diseases kill more people annually than war

Communities need rapid, affordable water safety assessment

The Solution

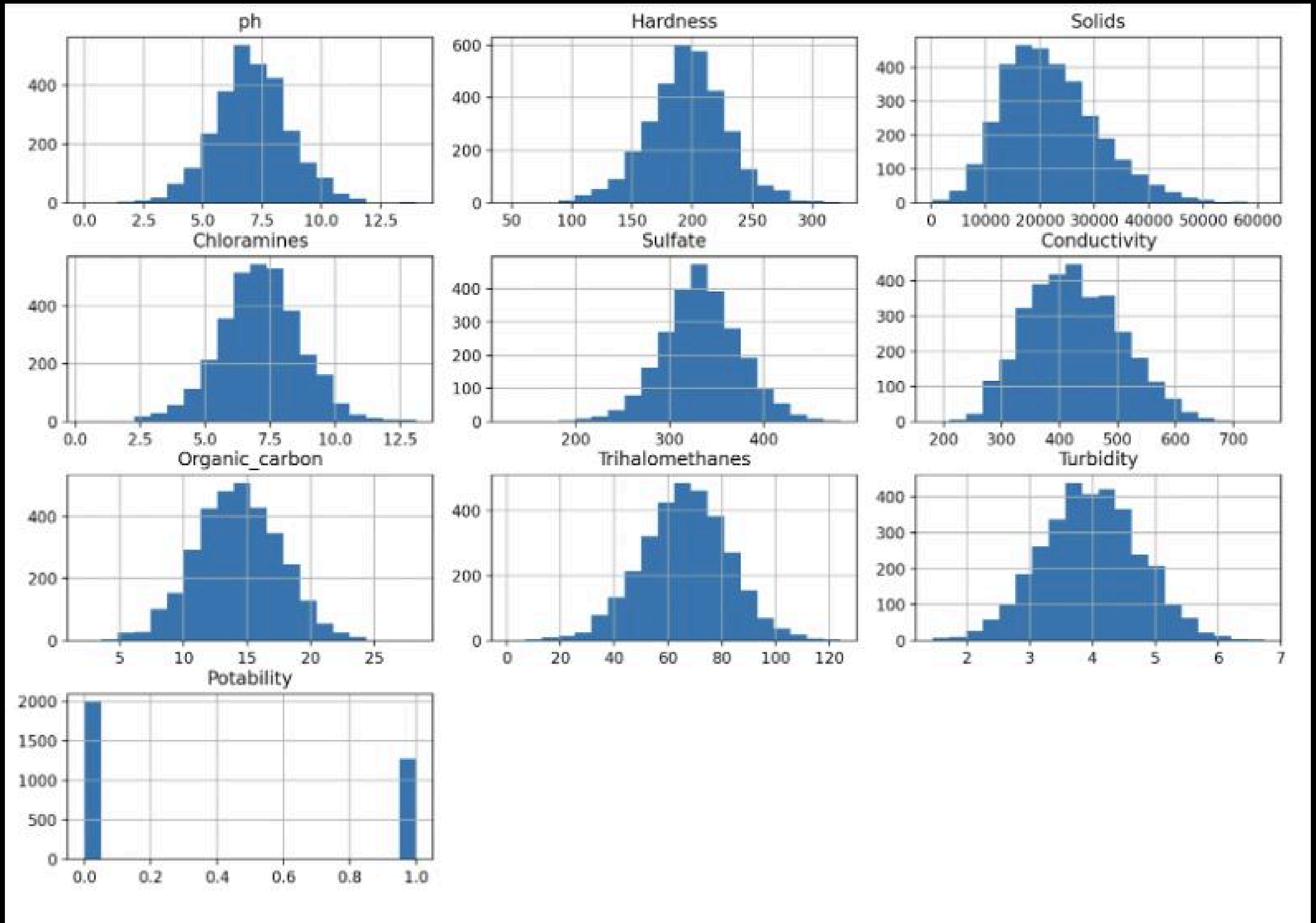
Machine Learning for Instant Water Quality Predictions

Dataset: 3,276 water samples with
9 parameters:

pH, Hardness, Solids, Chloramines, Sulfate, Conductivity, Organic Carbon, Trihalomethanes, Turbidity

Goal: Classify water as potable (safe) or not potable (unsafe)

Data Exploration




```
CORRELATIONS WITH POTABILITY:
Potability         1.000000
Solids             0.040674
Turbidity          0.022682
Chloramines        0.020784
ph                 0.014530
Trihalomethanes    0.009244
Hardness          -0.001505
Sulfate           -0.015303
Conductivity      -0.015496
Organic_carbon    -0.015567
Name: Potability, dtype: float64
```

Conclusion: Potability depends on complex, non-linear interactions

Model Selection

Model:	Cross Validation AUC:
Logistic Regression	**.48**
Random Forest	**.68**
Support Vector Machine	**.71**

Hyperparameter Tuning

Result: C=1.0, gamma='scale', RBF kernel

Broad Search

Tested kernels (RBF vs linear), C values, gamma

Refined Search

Narrowed C range, expanded gamma options

Fine-Tuning

Added class weights, final adjustments

Final Model Performance

Model is conservative

High precision, low recall

67%
Accuracy

70%
Precision

27%
Recall

.65
ROC-AUC

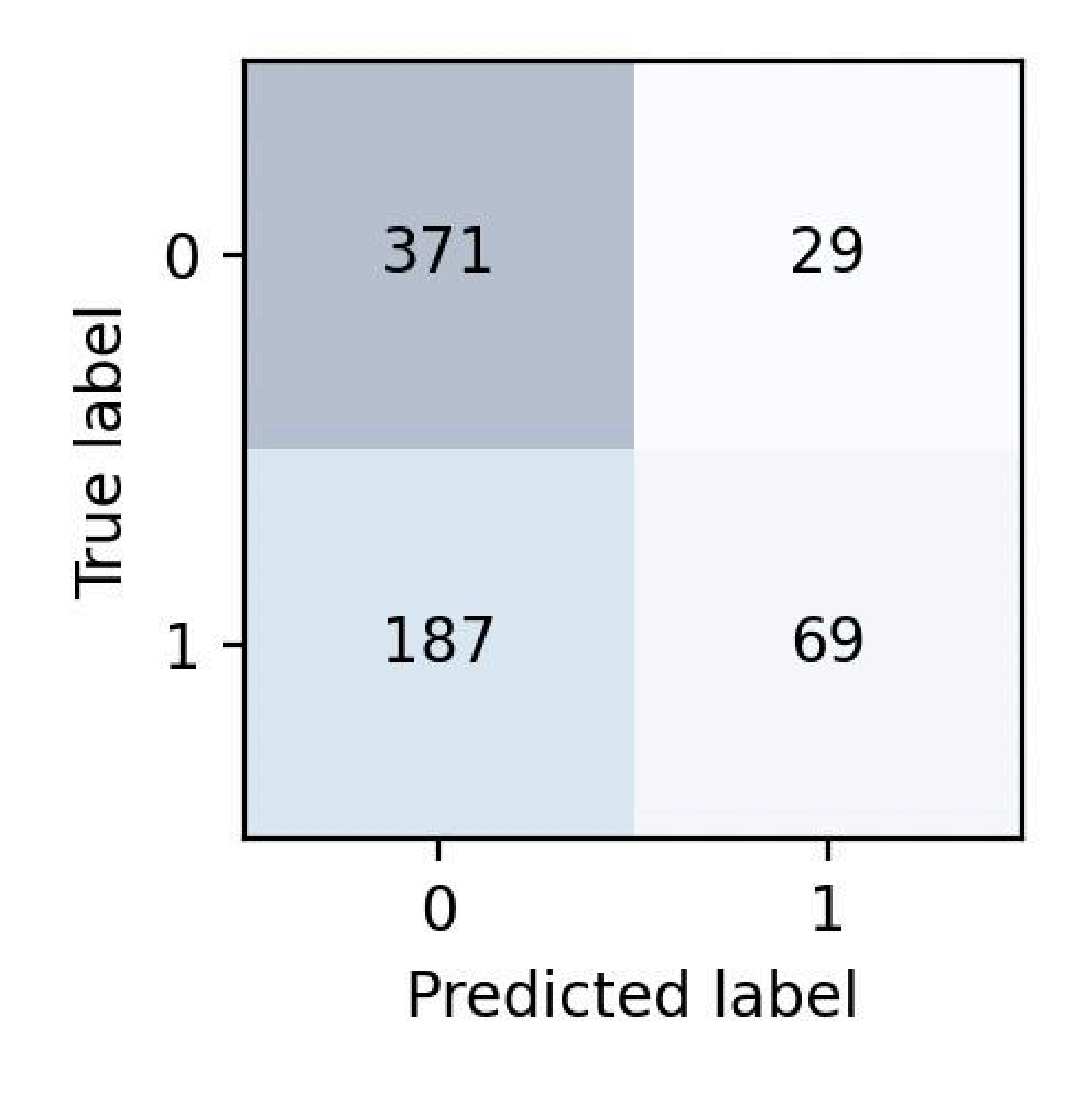
Confusion Matrix

371 True Negatives: **Correctly identified unsafe water**

69 True Positives: **Correctly identified safe water**

29 False Positives: **Unsafe water approved**
Dangerous

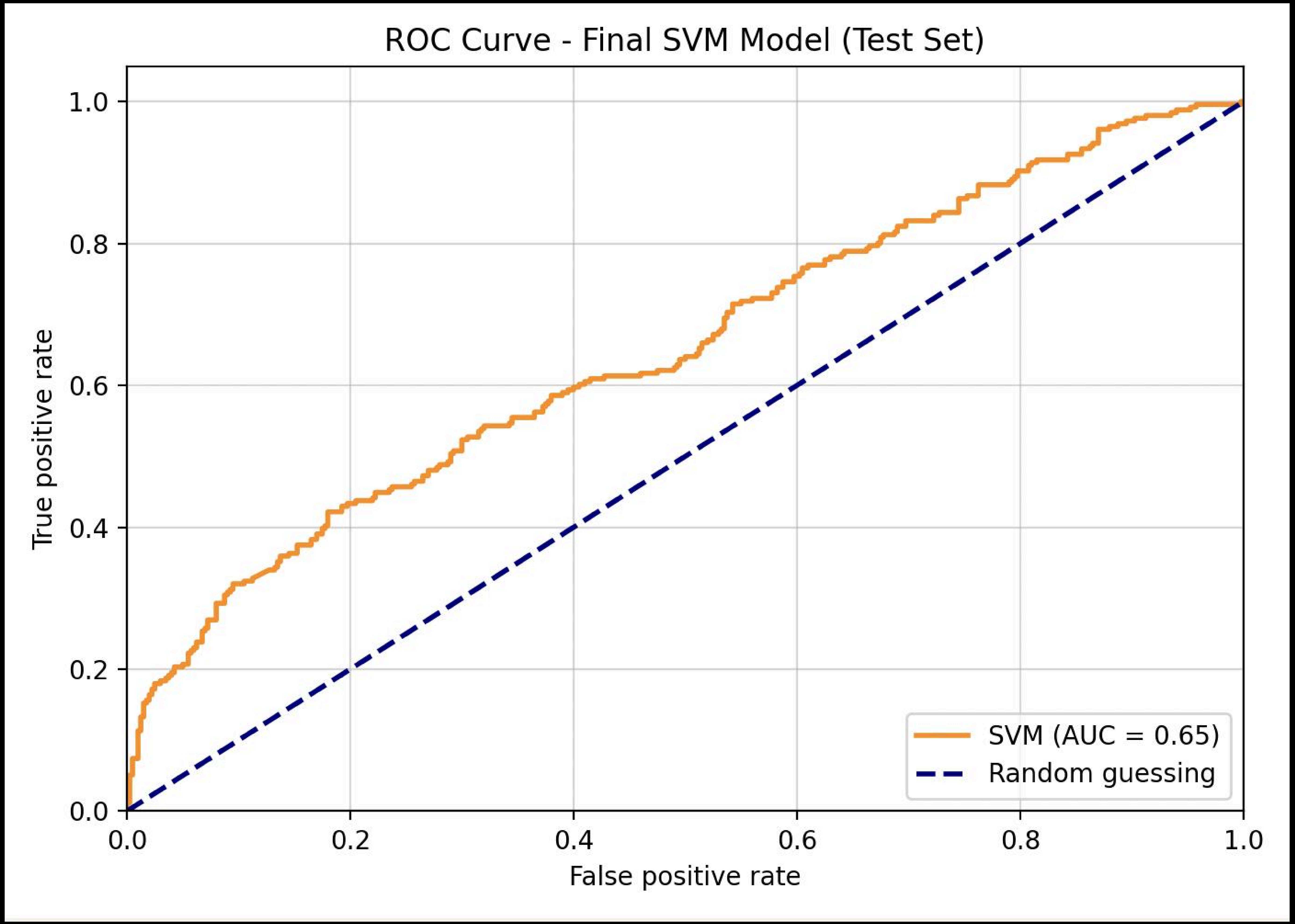
187 False Negatives: **Safe water rejected**
Cautious



ROC Curve

AUC = .65

Better than random guessing (0.50)
Room for improvement toward perfect (1.0)



Real World Application

Emergency Disaster Response

Instant water safety checks after hurricanes, floods, earthquakes

Refugee Camps

Affordable, regular monitoring of wells and local sources

Developing Regions

Aid workers assess sources without lab access

Take Aways

- ✓ Machine learning can predict water safety from chemical measurements
- ✓ Non-linear models (SVM) outperform linear approaches
- ✓ Conservative predictions prioritize safety
- ✓ Real-world potential: Rapid, affordable water testing for underserved communities

This technology could help ensure safe drinking water access worldwide